CAPSTONE PROJECT

AI STUDY BUDDY

Presented By:

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edunet foundation

OUTLINE

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT

- Students frequently struggle to understand complex educational concepts during self-study.
- Searching for answers online often yields overly long, confusing, or irrelevant results.
- Human teachers and tutors are not always available on-demand to provide immediate assistance or clarification.

PROPOSED SOLUTION

- Develop an AI-powered tool designed specifically for students to bridge the gap in self-study.
- Provide an on-demand "AI Tutor" to explain complex topics in simple, easy-to-understand terms.
- Include a "Smart Summarizer" to condense long, dense study notes into concise bullet points or short paragraphs.
- Feature an "Interactive Quiz" generator to dynamically create multiple-choice questions to test knowledge retention.

SYSTEM APPROACH

- **Programming Language:** Python.
- **Frontend Framework:** Streamlit (for building and sharing the web app quickly).
- **AI Integration:** OpenRouter API to connect with diverse, free Large Language Models (LLMs) like Gemini 2.0, DeepSeek, and Llama.
- **Environment Management:** python-dotenv for securely handling API keys.

ALGORITHM & DEPLOYMENT

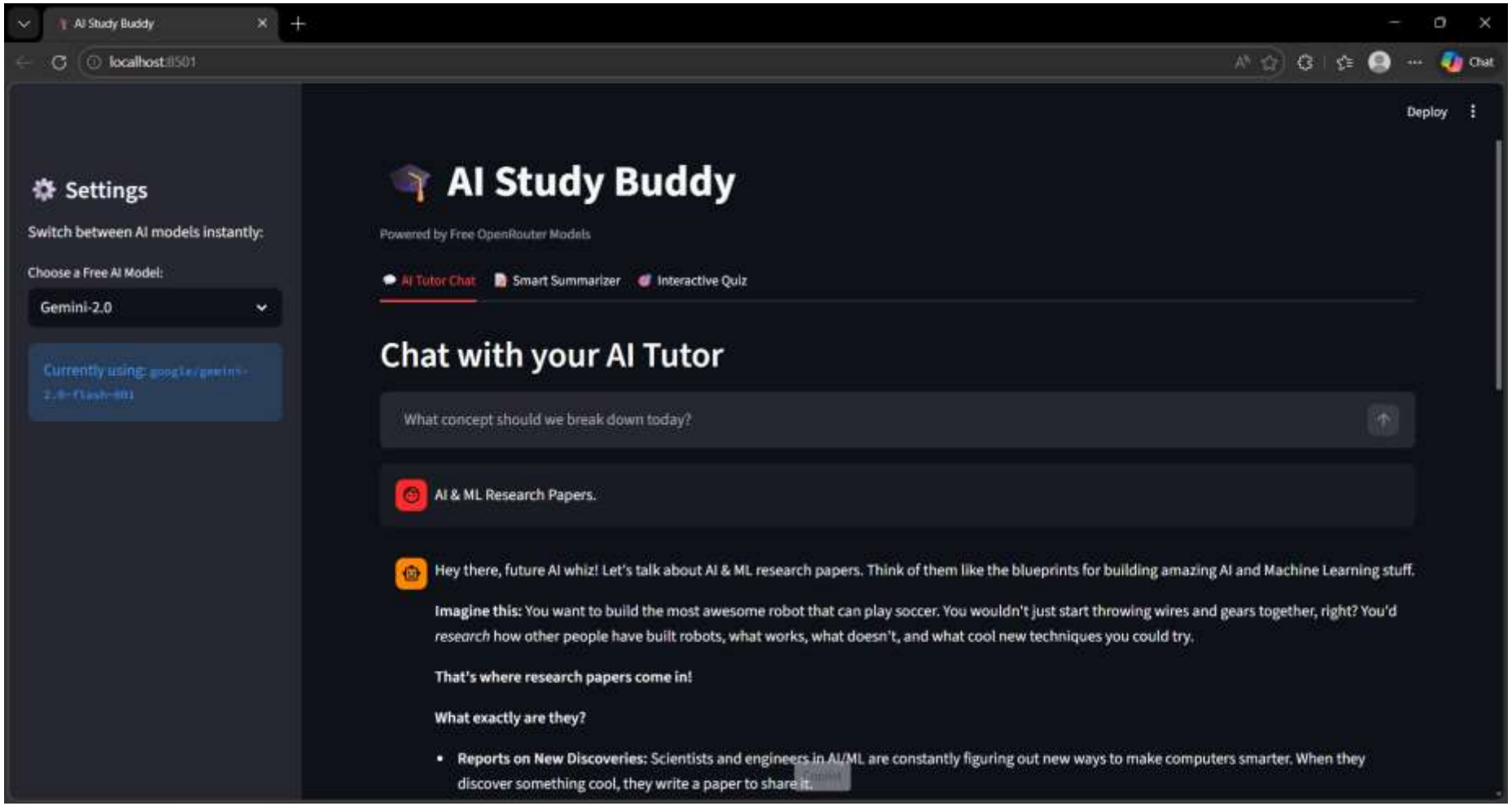
Algorithm:

- Uses Streamlit's built-in memory (session_state) to maintain chat history and track user quiz progress seamlessly.
- Dynamically routes user inputs to free AI models via the OpenRouter API using customized system prompts.
- Cleans the AI's responses using regular expressions to extract structured JSON data, allowing the app to auto-grade quizzes and give instant feedback.

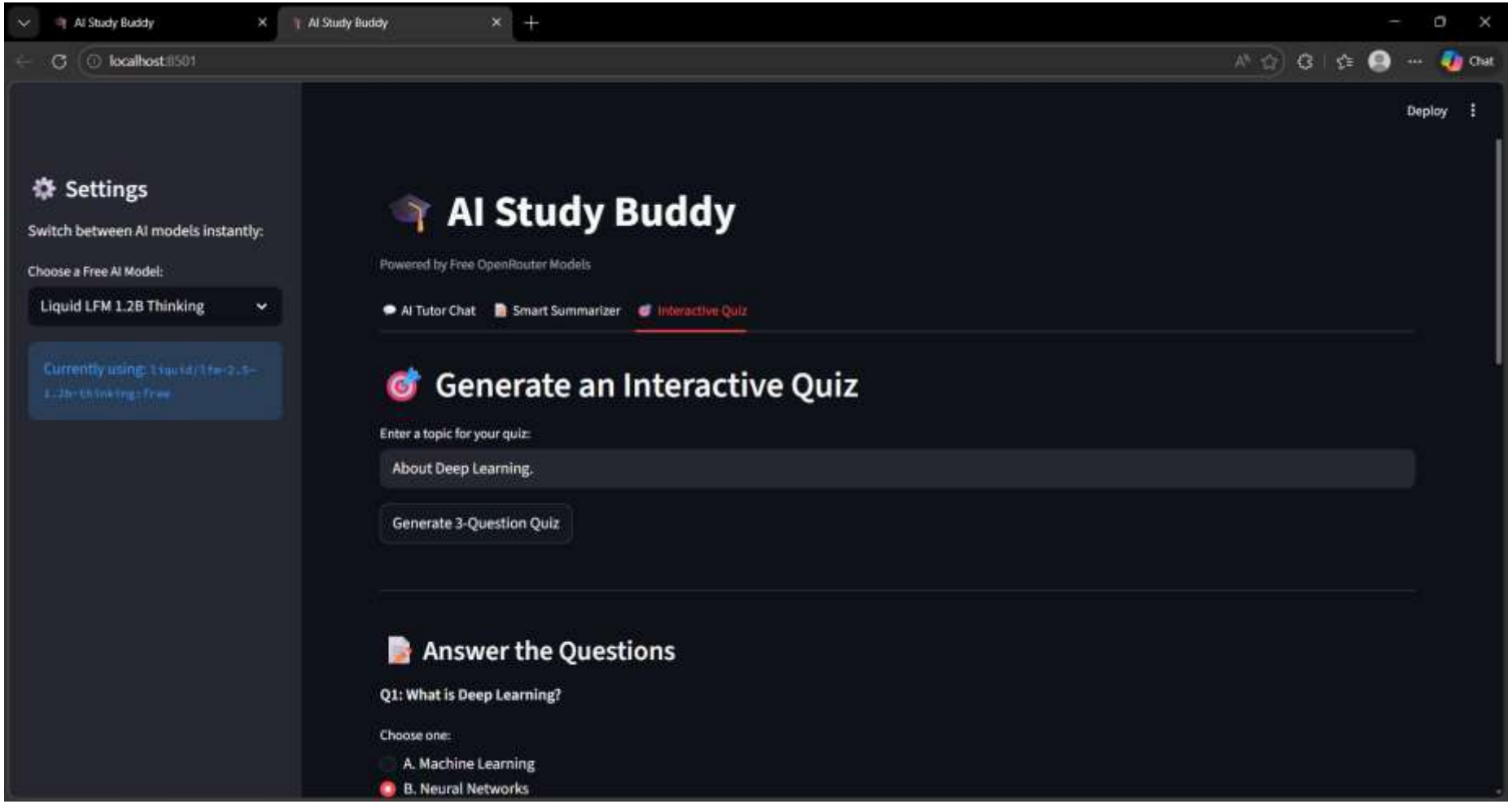
Deployment:

- Built entirely in Python using the Streamlit framework to quickly create an interactive, user-friendly web interface.
- Secures sensitive credentials (like the OpenRouter API key) using local environment variables (.env) to prevent unauthorized access.
- Designed to be easily deployed and shared publicly on the web using Streamlit Community Cloud.

RESULT


AI Study Buddy
localhost:8501
Chat
Deploy
Settings
Switch between AI models instantly:
Choose a Free AI Model:
Gemini-2.0
Currently using:
AI Study Buddy
Powered by Free OpenRouter Models
AI Tutor Chat
Smart Summarizer
Interactive Quiz
Chat with your AI Tutor
What concept should we break down today?
AI & ML Research Papers.
Hey there, future AI whiz! Let's talk about AI & ML research papers. Think of them like the blueprints for building amazing AI and Machine Learning stuff.
Imagine this: You want to build the most awesome robot that can play soccer. You wouldn't just start throwing wires and gears together, right? You'd research how other people have built robots, what works, what doesn't, and what cool new techniques you could try.
That's where research papers come in!
What exactly are they?
Reports on New Discoveries: Scientists and engineers in AI/ML are constantly figuring out new ways to make computers smarter. When they discover something cool, they write a paper to share it.

RESULT


AI Study Buddy
AI Study Buddy
localhost:8501
Chat
Deploy
Settings
Switch between AI models instantly:
Choose a Free AI Model:
Liquid LFM 1.2B Thinking
AI Study Buddy
Powered by Free OpenRouter Models
AI Tutor Chat
Smart Summarizer
Interactive Quiz
Generate an Interactive Quiz
Enter a topic for your quiz:
About Deep Learning.
Generate 3-Question Quiz
Answer the Questions
Q1: What is Deep Learning?
Choose one:
A. Machine Learning
B. Neural Networks

CONCLUSION

- The "AI-Powered Study Buddy" effectively solves the problem of information overload and tutor unavailability.
- By leveraging free OpenRouter models, the platform provides an accessible, highly capable educational assistant that makes learning more interactive and digestible.

FUTURE SCOPE

- **Flashcard Generation:** Expanding the tool to automatically generate printable or digital flashcards (as initially envisioned in the project description)

- **Document Uploads:** Allowing students to upload PDF textbooks or Word documents directly for summarization.

- **Progress Tracking:** Adding a database to save a user's quiz scores and chat history over time to track learning improvements.

REFERENCES

- **Streamlit:** A faster way to build and share data apps.
- **Link:** https://streamlit.io/

- **Python:**The core programming language used, including built-in modules like json and re.
- **Link:** https://www.python.org/

- **OpenRouter API:** The unified gateway used to connect to the various free Large Language Models.
- **Link:** https://openrouter.ai/

- **OpenAI Python SDK:** The library used to structure the requests and communicate with the OpenRouter API (from OpenAI import OpenAI).
- **Link:** https://github.com/openai/openai-python

- **Python-dotenv:**The library used to securely load your API keys from the .env file (from dotenv import load_dotenv).
- **Link:** https://pypi.org/project/python-dotenv/

GITHUB LINK

- Github Link : https://github.com/jjeelugu2005-debug/AI-Study-Buddy
- Deployment Link: https://ai-study-buddy-mgtghzmhkdyj7nyevb25zx.streamlit.app/

THANK YOU

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